FARMEXPRESS - BUSINESS MODEL CANVAS

Mobile-First Agricultural Marketplace Connecting Verified African Farmers with Diaspora Consumers & B2B Industry Buyers

React 19 | Express 5 | Neon DB | Paystack

1 KEY PARTNERS

Agricultural Supply Partners

- Verified smallholder & commercial farms across Oyo, Ogun, Kaduna, Kano, Benue & Ondo.
- Regional farming cooperatives & produce aggregation associations.

Logistics & Fulfillment

- 3PL carriers: FarmExpress Logistics, GIG Logistics, DHL Express, Kwik & local riders.
- Regional pickup hub operators for click-and-collect fulfilment.

Financial & Tech Infrastructure

- Paystack: card, bank transfer, USSD checkout, HMAC webhooks & automated API refunds.
- Vercel & Neon Serverless PostgreSQL for elastic cloud hosting & data integrity.

2 KEY ACTIVITIES

- Farmer & Business KYC Verification (CAC, National ID, Farm Permit, Tax ID review).
- Atomic stock reservation at checkout to prevent concurrent overselling.
- End-to-end order lifecycle management (PENDING -> PAID -> CONFIRMED -> SHIPPED -> DELIVERED).
- Multi-farm order consolidation & last-mile delivery tracking.
- Automated stale-order expiry (24h TTL) & Paystack refunds.

3 KEY RESOURCES

- Proprietary marketplace platform (React 19 SPA, Express 5 API, Prisma 6 ORM).
- Database of KYC-verified African farms & graded real produce catalog.
- DB-backed document & product image storage with immutable caching.
- Admin compliance suite with full security & operational Audit Log.

4 VALUE PROPOSITIONS

For Diaspora & Retail Buyers

- Direct access to authentic, farm-fresh African staples (Plantain, Cassava, Puna Yam, Cocoa, Ginger, Maize, Tomatoes).
- Instant stock reservation, transparent NGN pricing & live delivery tracking.

For Industry & Bulk Buyers

- Verified bulk sourcing from audited farms & cooperatives for food processing and export.
- Reliable quantity availability with platform-coordinated multi-farm fulfilment.

For Verified Farmers

- Direct reach to higher-margin diaspora, urban & industrial buyers without middlemen.
- Simple mobile dashboard for product photos, stock control & order dispatch.

5 RELATIONSHIPS

- Verified farm badges & transparent KYC trust signals on every listing.
- Real-time in-app, email & SMS notifications at every order milestone.
- Verified-buyer product ratings & reviews unlocked after order delivery.
- Admin-backed refund guarantee & dispute resolution.

6 CHANNELS

- Mobile-first web storefront with instant search & category filtering.
- Role-tailored dashboards for Buyers, Farmers & Platform Admins.
- Direct Home Delivery (doorstep rider dispatch) & Regional Pickup Hubs.
- Automated transactional alerts (In-App Notification Bell + Email/SMS webhooks).

7 CUSTOMER SEGMENTS

1. Diaspora Buyers (BUYER)

- Africans abroad purchasing fresh farm produce for family households or export packages.

2. Domestic Urban Buyers (BUYER)

- Households ordering farm-fresh fruits, tubers, grains & vegetables online.

3. Industry & B2B Buyers (INDUSTRY)

- Food processors, agro-allied manufacturers, hotels & commodity exporters ordering in bulk.

4. Verified Farmers (FARMER)

- Smallholder & commercial farmers seeking digital storefront & guaranteed payments.

8 COST STRUCTURE

Platform & Fulfilment Costs

- Payment Gateway Fees: Paystack processing (1.5% + NGN 100 per local transaction, capped at NGN 2,000) & refund handling.
- Logistics & Hub Operations: Last-mile carrier dispatch payouts, pickup hub management & produce quality checks.
- Cloud & Messaging Infrastructure: Vercel serverless compute, Neon PostgreSQL storage, Email & SMS notification webhooks.
- Compliance & Operations: Farmer KYC verification reviews, multi-farm order coordination & customer support.

9 REVENUE STREAMS

Transaction & B2B Monetization

- Marketplace Commission (Take-Rate): 8% - 12% commission on Gross Merchandise Value (GMV) of completed farm orders.
- B2B & Industry Bulk Fulfilment Fee: 5% - 8% aggregation and quality-assurance margin on multi-farm industrial orders.
- Delivery & Logistics Margin: Service fee on doorstep delivery and express cold-chain coordination.
- Value-Added Services (Expansion): Featured farm listings, export grading certification & working-capital financing.

Confidential - Product & Strategy Overview

A. END-TO-END ORDER, FULFILMENT & PAYMENT LIFECYCLE

1. PENDING
Buyer places order
Stock reserved atomically

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2. PAID
Paystack / Offline
Webhook + redirect verified

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3. CONFIRMED
Farm accepts
Harvest & packing started

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4. SHIPPED
Carrier dispatched
Tracking ID & ETA sent

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5. DELIVERED
Buyer receives
Verified reviews unlocked

Safeguards: (1) Unpaid PENDING orders auto-expire after 24h and restock inventory. (2) Cancelled paid orders trigger direct Paystack API refunds with full Audit Log.

Multi-farm Orders: Single-farm orders are fulfilled directly by the owning farmer; mixed-farm orders are coordinated by FarmExpress Operations.

B1 PLATFORM ROLE MATRIX

Buyer & Industry Buyer

- Browse live produce catalog with search, category filter & pagination.
- Reserve stock at checkout with Home Delivery or Pickup Hub selection.
- Pay via Paystack hosted checkout (Card, Transfer, USSD).
- Track live shipment timeline & review delivered produce.

Verified Farmer

- Upload KYC documents (Farm Permit, CAC, National ID, TIN).
- Create listings with direct device photo upload & stock counts.
- Confirm paid orders, attach carrier tracking numbers & mark delivered.

Superadmin (Demola Blaze)

- Review uploaded KYC documents and approve/reject with notes.
- Oversee all orders, mark offline payments & issue Paystack refunds.
- Inspect immutable platform Audit Log & live revenue analytics.

B2 TARGET UNIT ECONOMICS

Retail & Diaspora Basket (Sample)

- Average Order Value (AOV): NGN 35,000
- Platform Take-Rate (10%): NGN 3,500
- Delivery Coordination Margin: NGN 1,000
- Paystack Processing Fee (~1.5% + NGN 100): -NGN 625
- Net Contribution per Order: ~NGN 3,875 (11.1% of GMV)

Industry / B2B Bulk Order (Sample)

- Average Bulk Order Value: NGN 450,000
- B2B Aggregation Fee (7%): NGN 31,500
- Paystack Fee (Capped at NGN 2,000): -NGN 2,000
- Quality & Dispatch Ops Cost: -NGN 4,500
- Net Contribution per B2B Order: ~NGN 25,000

Key Operational SLAs

- KYC Verification Review Turnaround: < 12 hours
- Unpaid Stock Reservation Lock (TTL): 24 hours
- Farm Dispatch Confirmation after Payment: < 24 hours

B3 PRODUCT CATALOG & TECH STACK

Core Verified Produce Categories

- Fruits: Fresh Green & Ripe Plantain Bunches.
- Roots & Tubers: Fresh Cassava Tubers, Puna White Yams.
- Grains: Yellow Maize, Raw Fermented Cocoa Beans & Pods.
- Vegetables & Spices: Roma Tomatoes, Dried Red Chili, Split Ginger.
- Legumes: Raw Shelled West African Groundnuts.

Production Technology Stack

- Frontend: React 19, TypeScript, Vite 7, Tailwind CSS 4.
- Backend API: Node.js, Express 5, Zod 4 validation, JWT + bcrypt.
- Database: Neon Serverless PostgreSQL + Prisma 6 migrations.
- Quality & CI: Automated API test suite + GitHub Actions CI.
- Deployment: Full-stack Vercel (Frontend + /api serverless) & Render.